

ε-greedy Algorithm.

Calculation:-

Mathematical Formula:

$$Q_{n+1}(a) = Q_n(a) + \frac{R - Q_n(a)}{N(a)}$$

- $Q_n(a)$ = current value of a machine a .
- R = newly obtained reward.
- $N(a)$ = number of times Machine a has been selected after the agent play.
- $Q_{n+1}(a)$ = updated estimated value.

Initially,

$$N(A) = N(B) = N(C) = 0$$

$$Q(A) = Q(B) = Q(C) = 0$$

Calculation for play 1

ε-greedy algorithm

Selects Machine B,

$$R = 6$$

Before Play, $N(B) = 0$

$$Q(B) = 0$$

After selecting B, at machine

$$N(B) = 0 + 1 = 1$$

Using the incremented formula

$$Q(B) = Q_1(B) + \frac{R - Q_1(B)}{N(B)}$$

Sub values

$$Q(B) = 0 + \frac{6 - 0}{1}$$

$$Q(B) = 6$$

th. after play 1

Machine

	A	B	C	Q
A	0	0	0	0.00
B	1	6	0	6.00
C	0	0	0	0.00

	A	B	C	Q
A	0.00	0	0	0.00
B	0.00	6.00	0	6.00
C	0.00	0	0	0.00

Calculation for play 2:

$$R = 7$$

Before play 2: $N(B) = 1$
 $Q(B) = 6$

After selecting B:

$$N(B) = 1 + 1 = 2$$

Formula:

$$Q(B) = Q(B) + \frac{R - Q(B)}{N(B)}$$

Ex.

$$Q(B) = 6 + \frac{7-6}{2}$$

$$Q(B) = 6 + 1/2$$

$$Q(B) = 6.5$$

Normal
avg:

$$Q(B) = \frac{6+7}{2}$$

$$Q(B) = \frac{13}{2} = 6.5$$

So, after play 2.

Machine	N	R	Q
A	0	0	0.00
B	2	15	6.50
C	0	0	0.00